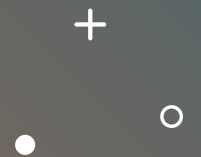


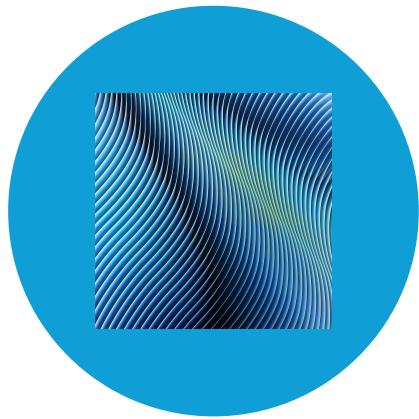


Judgment and Prediction (Part 1)

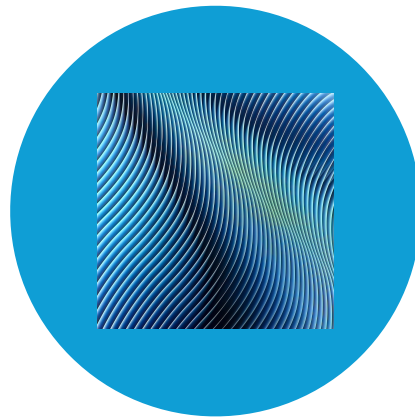
Is anyone good at it?



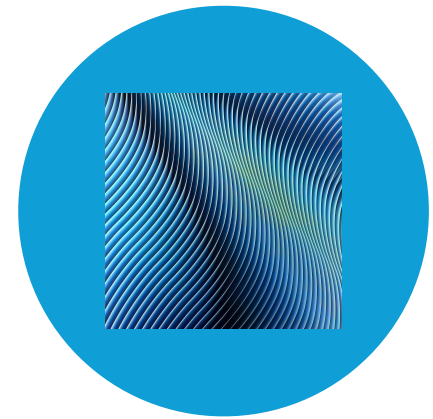
The Wisdom of Crowds



WHY LOOK AT
CROWDS?

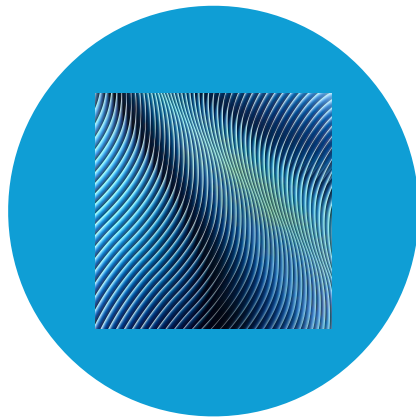


AREN'T THERE
ALTERNATIVES?

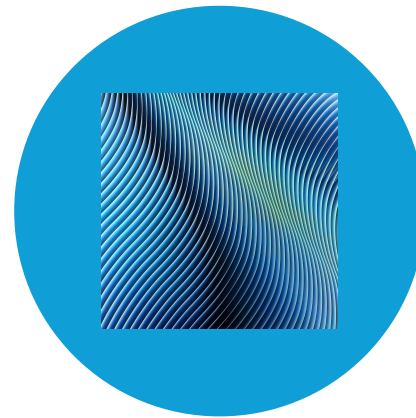


ARE THEY ANY
BETTER?

The Wisdom of Crowds

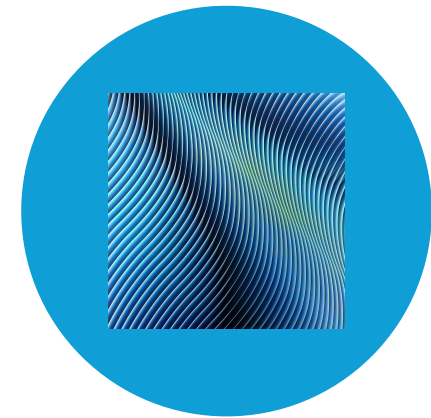


WHY LOOK AT CROWDS?



AREN'T THERE ALTERNATIVES?

YES THERE ARE.

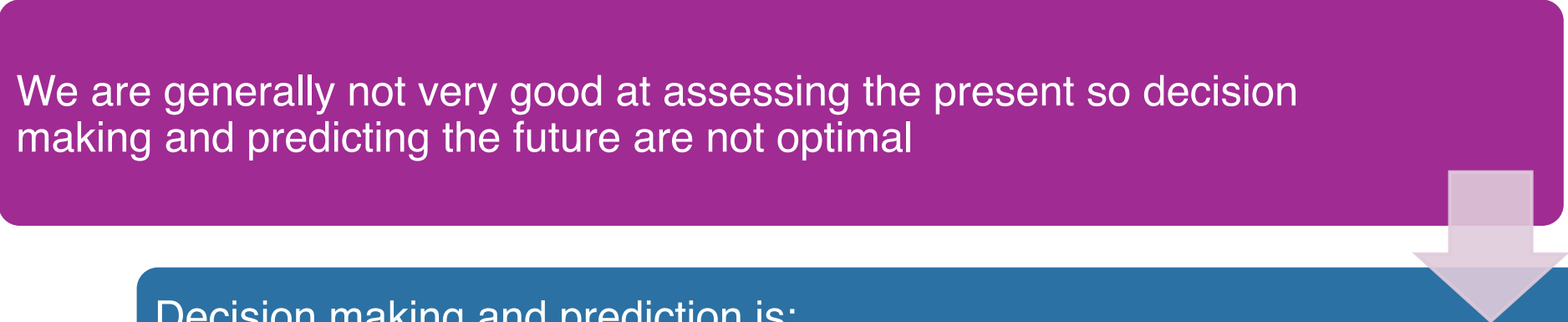


ARE THEY ANY BETTER?

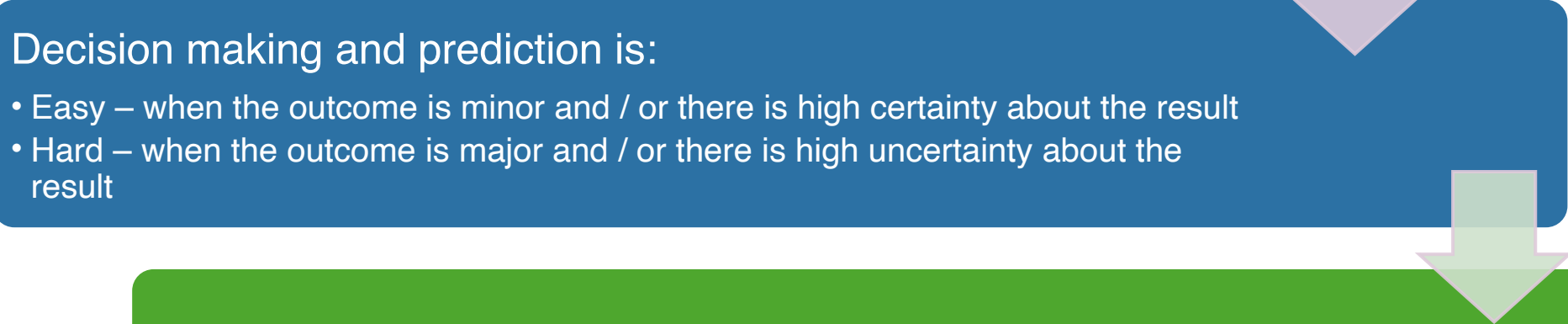
OFTEN, THEY ARE NOT

Introduction

We are generally not very good at assessing the present so decision making and predicting the future are not optimal

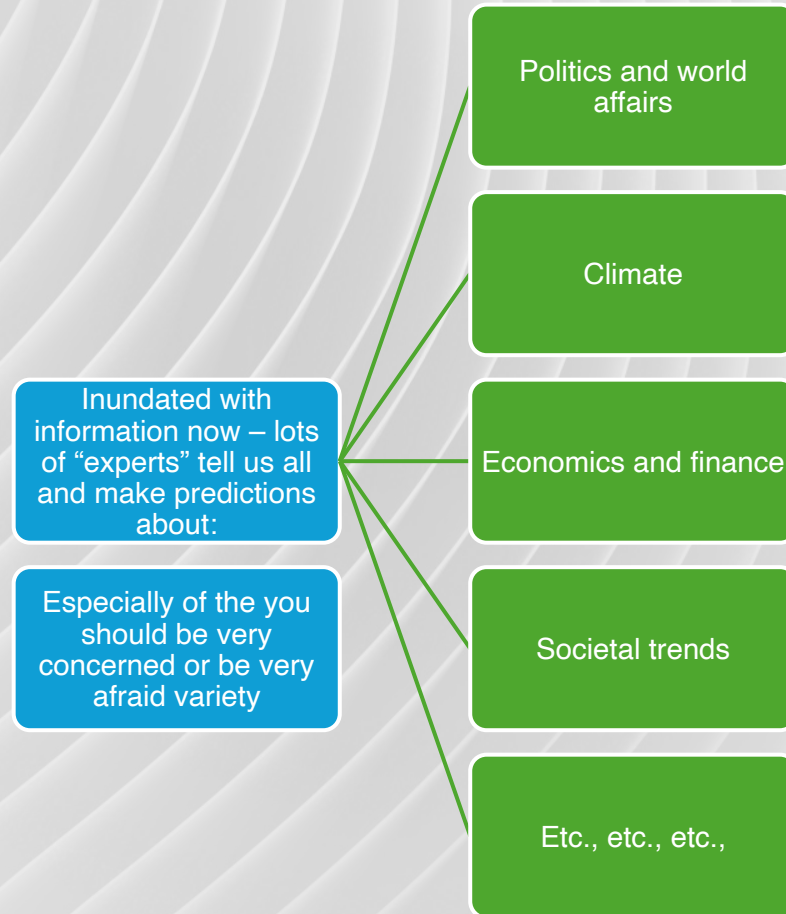


Decision making and prediction is:

- Easy – when the outcome is minor and / or there is high certainty about the result
 - Hard – when the outcome is major and / or there is high uncertainty about the result
- 

As an alternative to self reliance, and before we get to crowds, we will look at experts – how good are they?

Experts are Everywhere



Experts are Everywhere, But ...

Since we want to know, they gladly address probably our biggest questions – what is happening and what happens next



But do they provide information and guidance or just noise?



Who are they? Why are they the ones we hear?
How good are they? Are they better than us? Are they the best we can do?

What This Presentation Is and Is Not About

- Parallel issues to this presentation include:
 - The internet and social media
 - All those persons claiming to be an expert or to have asked experts
 - Misinformation and Disinformation
 - AI
- These issues are related and could arise in discussion but are *not* our topic today
- Our focus is on people considered to be *bona fide* experts, people who have real knowledge and experience in an area

Sources

- One source on Experts

Sources

- One source on crowd wisdom

Sources

Introduce books

- Expert Political Judgment
- Wisdom of Crowds

First book: dense book evidencing results of a multi-year study started 40 years ago – never heard of it but it is the genesis of things you may have heard:

- The idea that experts are no better than dart throwing monkeys
- Group or crowd theory, including prediction markets

Second book – you will need to stay tuned for it next month



Expert Political Judgment

- The project – to ascertain who is good at judging and predicting and, more importantly WHY and HOW
- They chose an **extremely** complex subject area – political and economic judgment and forecasting
 - This was initially in the 1987 – 1992 so they focused on topics germane at the time, such as political and economic events in Russia, Saudi Arabia, South Africa, Japan, India, Canada, Poland, Nigeria, etc.
- Who conducted the study?
 - Psychologists, who were studying human behavior
 - Not Political Scientists, Historians, Etc.

Wait Just a Minute Here!



- Before attacking experts, remember how vital they are and remember a key distinction:
 - short term predictions with easily quantifiable predictions that provide feedback for improvement vs.
 - long term predictions with harder to quantify outcomes and vaguer and more remote feedback
 - Former: doctors, fire fighters, air traffic controllers, weather forecasters – they do quite well
 - Latter: world or economic events where larger real-world judgments are made and where real or even extreme uncertainty exists – they don't do so well

Warning(s)

This discussion is a very brief and simplified summary of the complexities of the study described in the book.

- But the tests and results are fascinating (to me anyway)
- Will try to focus on the forest and not get bogged down in the trees

Subject area required researchers to be nimble and sensitive in crafting the questions and analyzing the answers

Overview (1)

- So what exactly were they “measuring” or studying?
 - **Good judgment**, which they defined as:
 - The fit between internal mental representations and correspondent properties of the outside world
 - So good judgment is shown by those who see the world as it is or soon will be
 - Extra credits for those whose see it far in advance of the rest of us
 - Extra demerits for those who fail to see it long after the rest of us do

Overview (2)

- Good judges, who should both get it right and think the right way to be accurate
- Two types of tests:
 - **Correspondence tests**
(empirical) – how well do private beliefs map onto the real world
 - **Coherence & process tests**
(logic) – are beliefs internally consistent and updated with new information (including Bayesian updating)

Overview

(3)

- Assessing *judges* can be made (at least) somewhat unpolitical, assessing *judgment* cannot; e.g., how confident can we now be that the optimal outcomes resulted from:
 - Truman's dropping the bomb,
 - Kennedy's handling of Cuba,
 - Bush's WMD inspired invasion of Iraq
- these are inherently political judgments (and not their or our subject)

Overview (4)

- But devising a testing system for ascertaining good judgment is a difficult process:
 - How to score subjective probability forecasts?



A Little More Detail – We Look at the Trees

- Also measured (on 9-point scale of agreement-disagreement):
 - difference in ideological orientation (asking e.g., I believe that balance of power politics remains the reigning principle in world politics)
 - styles of reasoning (asking e.g., I usually make important decisions quickly and confidently; and asking them to self-sort themselves into being (more of) a fox or hedgehog)
 - Most importantly, they also measured cognitive styles (discussed below)
- 

Just Some of Their Challenges

- In measuring how well did a person “get it right”, what about
 - unlevel playing fields → uneven and unfair results (some questions easier than others)
 - measuring hits risks ignoring they were purchased by too many false alarms (misses)
 - properly balancing value of hits and false alarms
 - challenge of scoring subjective probability forecasts (avoid weasel words like maybe or likely) in favor of probabilistic forecasts (assign a value: 0-100% for outcomes with all possible outcomes totaling exactly 100%)
 - defining reality – what “really” did end up happening can be ambiguous

Study Participants (1)

- Multi-year studies (1987-1988 continuing until 2013)
 - Initially seeking predictions of outcomes between 1998 and 2003 but later added more
 - Who were the experts that agreed to participate in this study?
 - 284 persons initially
 - 76% male
 - average age = 43
 - average relevant work experience = 12.5 years
 - highly educated; 52% doctoral degrees; 96% post graduate training

Study Participants (2)

- from disciplines relevant to the study:
 - international relations
 - economics
 - national security & arms control
 - journalism
 - diplomacy
 - International law
- Working in
 - academia
 - government
 - think tanks & foundations
 - international institutions
 - private sector, including media
- other background indicia of (recognition at least) expertise
 - 61% interviewed at least once by a major media outlet
 - 21% - interviewed more than 10 times
 - 80% had at one time served as formal or informal consultants on international political or economic issues to government, private sector, international agencies or think tanks

Skepticism of the Ability to Forecast (1)

They recognized and anticipated skepticism as to whether you can measure what they were trying to measure: the accuracy of predictions

Skeptics would argue that linking the past and future are instead:

- The result of path dependency – e.g., the “rise of the West” became inevitable and irreversible at some point
- Governed by complexity theory with chaos rumbling through incomprehensible channels – e.g., tipping points
- Explained by game theory – e.g., rational agents second guessing one another
- Or by probability theory – e.g., events have multiple causes so that assessing what happened makes predicting what will happen look far easier than it is (e.g., examining vs. predicting plane crashes or earthquakes)

Skepticism of the Ability to Forecast (2)

—

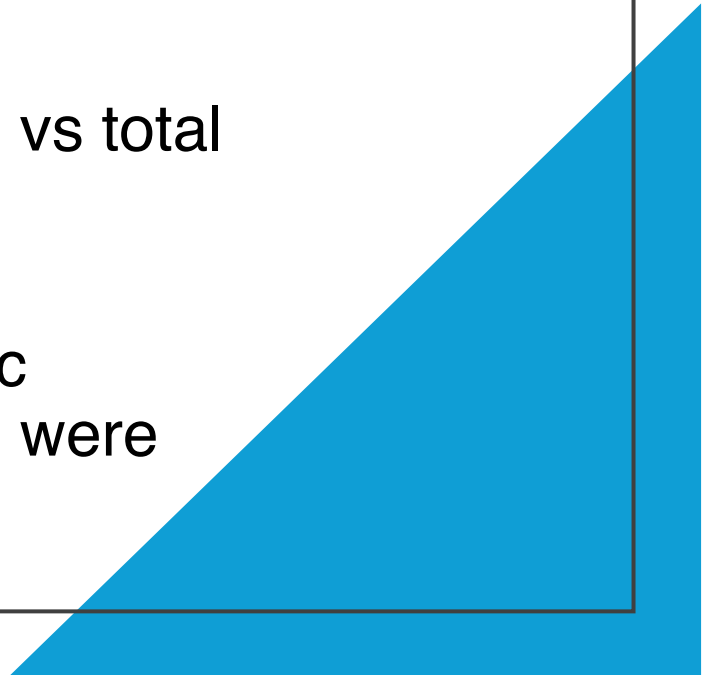
- People are inherently bad at predicting because we:
 - prefer simplicity and are poor at identifying relevant combinations of antecedents with resultant consequences
 - Are averse to ambiguity and dissonance, with a disdain for consideration of counterfactual scenarios that test our assumptions
 - want to be in control and believe that events are predictable, not capricious
 - disdain the idea of randomness and unpredictability as shown by our aversion to probabilistic strategies that recognize the existence of error

Testable Skeptical Arguments – That Experts Have Little to Add

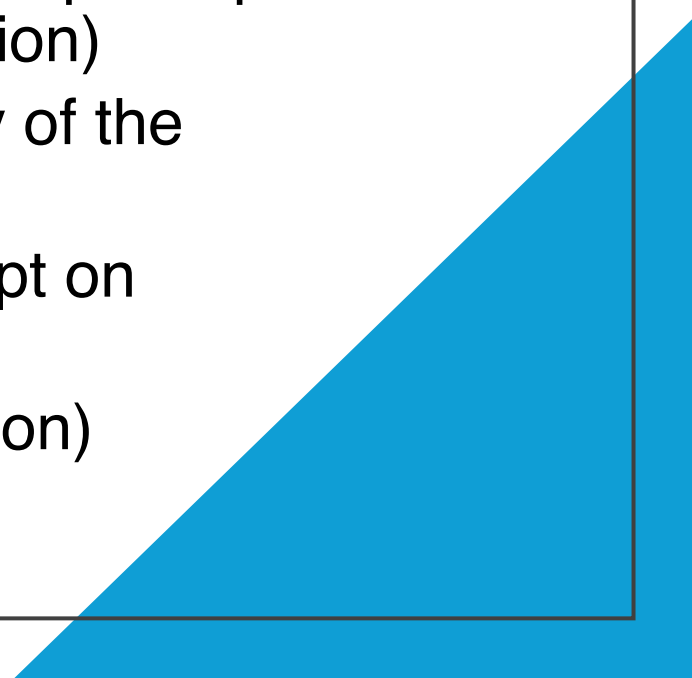
- Debunking - in turbulent times when advice is most valuable, random prediction generators (dart throwing monkeys) are as good as experts
- Even if correct, experts will see diminishing rates of accuracy going forward
- 15 minutes of fame does not equate to overall forecasting accuracy in other areas
- Loquaciousness (hot air balloon) or expertise (knowledge) + ability to fashion elaborate arguments → overconfidence → diminishing returns
- Seduced by fame, experts will be tempted to offer quotable sound bites
- The illusion of sustainable consistency over time

Results – the Measures of Accuracy

- Results - the evidence of 27,451 predictions measuring:
 - **Calibration** – predicted frequencies of outcome vs. objective (actual) frequencies
 - **Discrimination** – predicted variation in outcome vs total variation*
- *They were *usually* required to make probabilistic predictions allowing for various outcomes, which were thus measurable.

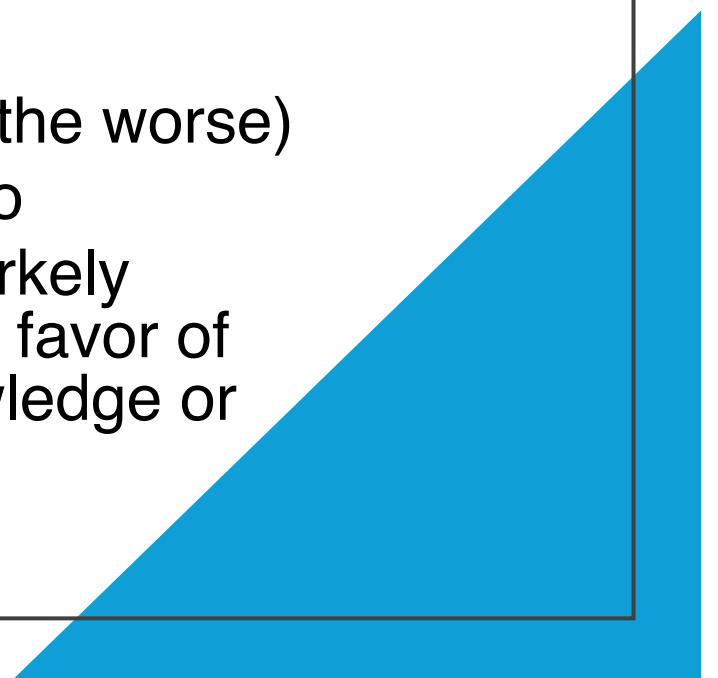


Results re: Skeptical Arguments (1)

- Debunking – experts beat chimps on discrimination but not on calibration
 - Best predictors – several algorithmic models (except simple base rate continuation which allows for no variation)
 - Middle – human experts, who could not best any of the sophisticated models
 - Worse – random generators (the “chimps”) except on calibration
 - Worst – simple base rate continuation (no variation)
- 

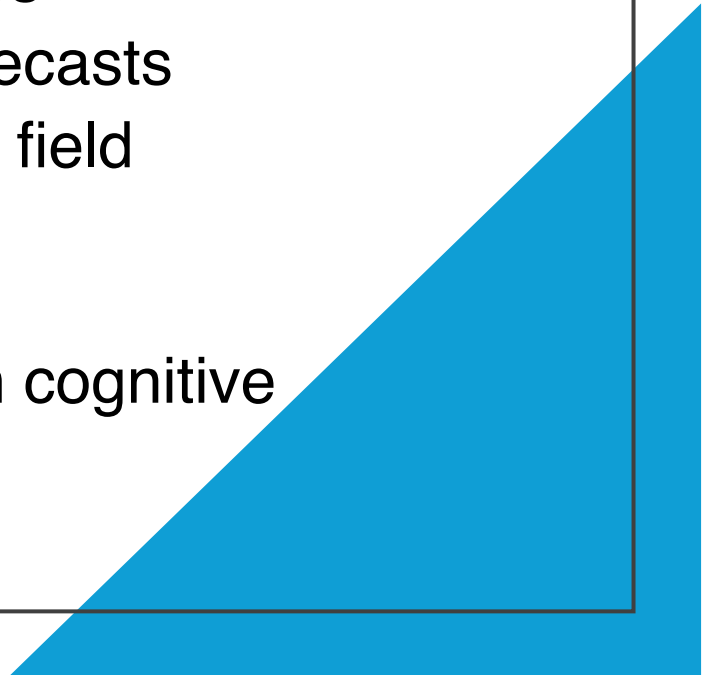
Results re: Skeptical Arguments (2)

- Diminishing returns –
 - experts in their field did **NOT** do better than dilettantes (experts in another field) despite any differences in education, experience, access to classified information
 - their errors were consistent and both
 - overestimated change (for the better and for the worse)
 - underestimated continuation of the status quo
 - but the worst predictors were “briefly” briefed Berkely undergraduates (so when rejecting algorithms in favor of humans, don’t rely on those with little base knowledge or incentive to be right)



Results re: Skeptical Arguments (3)

- 15 minutes of fame – here the focus really shifts to individuals from the aggregate, and we see something different appearing;
- **some forecasters really are better**, they:
 - have higher calibration AND discrimination scores
 - are better at both short term and longer-term forecasts
 - are better both within their field and outside their field (dilletantes)
- A hint at what is ahead: they are better based on cognitive style



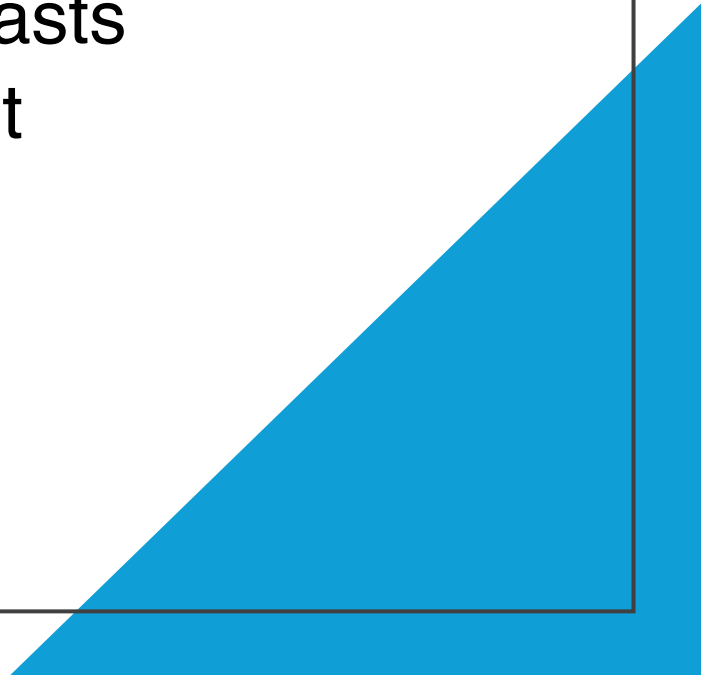
Results re: Skeptical Arguments (4)

- Hot air balloon – as expertise rises, so does confidence.
- Thus while both experts and dilettantes overestimated change (especially for the worse), the experts overestimated by more and made more big errors
 - Greater expertise → more elaborate justifications for their predictions and
 - The more lopsided the arguments they made, the more they supported their forecast



Results re: Skeptical Arguments (5)

- Factors independent of the experts per se
 - Seduction – experts in demand were indeed more overconfident (supply) in their forecasts
 - Indefinitely sustainable illusion – a subtest showed that belief in reliance on experts increases when the stakes are higher (demand)



This Looks Bad for Experts, Right?

- *Overall*, all experts were overconfident in accurately predicting the future
- *In the aggregate*,
 - Sophisticated algorithms beat the experts
 - experts beat the chimps (random projections) – but only by a slim margin
 - experts in their own field did **NOT** beat sophisticated dilettantes

Have Experts *Nothing* to Offer?

So it would appear.



BUT WAIT, the rest of the story IS the story;

some forecasters did do
much better than
others,

who and why is the
story here



Who Did Better?

- What do you think?

Who Did Better - What Did Not Matter?

-
- What did **NOT** not correlate with predicting accuracy (calibration)
 - Professional background –
 - Education, years of experience, access to classified information, type of work
 - **Except** – contact with the media was **negatively** correlated with accuracy
 - Ideological orientation (worldview) **except** when cognitive style started to show a bit; so that moderates were better than:
 - those oriented on the left or the right
 - those identified realists or idealists
 - boomsters (future is bright) and much better than doomsters (future is bleak)
 - So, extremism was a negative factor for forecasting

Description of a Key Differentiator*

- **Hedgehog**: thinkers who know one “big thing,” aggressively extend it into new domains, display bristly impatience with those who don’t “get it,” and express considerable confidence that they are pretty good forecasters - (ironically) at least in the long term (as in a “I’ll eventually be right” or “I almost was right” vein). Hubris?
- **Fox**: thinkers who know many small things (including tricks of their trade), are skeptical of grand schemes, see explanation and prediction not as deductive exercises but rather as flexible “as hockery” that require stitching together diverse sources of information, and are diffident about their forecasting prowess. Humility?
- **These are the extremes; we all fall somewhere between the two.**

*A quality or feature that makes something different from, and usually better than, other similar things.

Hedgehogs and foxes are terms coined by Isaiah Berlin, a social and political theorist, philosopher, and historian of ideas

Who Did Better - What Did Matter (1)?

-
- Cognitive style:
 - Moderates were more likely to be foxes
 - Foxes were the higher scorers, Hedgehogs were the lower scorers, Foxes dominated on
 - Calibration and discrimination
 - In a generalized ways (experts vs. dilettantes, moderates vs. extremist, short term vs. long terms forecasts)
 - As well as stable vs. turbulent outcomes
 - **BUT** – only the very best foxes came *even close* to besting sophisticated extrapolation algorithms
 - And the worst experts were truly dismal

Who Did Better - What Did Matter (2)?

-
- Worst performers were hedgehogs extremists making long term projections within their domain of expertise
 - Best performers were foxes making short term projections within their domain of expertise
 - The complex interplay of style and other effects:
 - Increased knowledge resulted in lower accuracy for hedgehogs because
 - They have a greater need for closure and desire to avoid ambiguity
 - Greater knowledge gives them access to more supportive ammo and makes it easier to trivialize opposing information
 - Similarly, hedgehogs performed worse on (inherently more unpredictable) longer range forecasts

What Could All Experts Have Done Better

**Use processes
that improve
judgment and
prediction**



Consideration of alternatives by asking:

- If my views on the current circumstances are correct, how like is the result I don't predict
- If my views are incorrect, how likely are other results,

Updating views and predictions based on changed conditions – Bayesian updating

In making their predictions, most experts failed to use either process and instead clung to old ideas too long – unwillingness to admit mistake

Conclusions (1)

- ***What*** experts think and know matters far less than ***how*** they think (e.g. knowledge and ideology are not relevant but flexibility is)
 - (Isaiah Berlin's) Foxes outperform his Hedgehogs
 - Fox – knows many little things, draws from an eclectic array of traditions, and accepts ambiguity and contradictions as inevitable
 - Hedgehog – knows one big thing, toils devotedly within one tradition, and reaches for formulaic solutions to ill-defined problems
 - Resulting in a double irony – a perversely inverse relationship between the indicators of good judgment and the qualities that the media prizes in pundits (the tenacity to prevail in ideological combat)

Conclusions (2)

- Beyond the aggregate,
 - those most sure of their big picture views (hedgehogs) performed worse than
 - those who stuck to the data and considered other viewpoints (foxes)
 - for all forecasts but especially for longer range forecasts
- But beware of sweeping generalizations; i.e., hedgehogs are not always wrong

Conclusions (3)

- Composite view of good judgment emerged: forecasting accuracy linked to both:
 - curiosity, open mindedness, and an unusual tolerance for dissonance
 - awareness of the fragility of their own forecasting achievements (humility?)
- Thus, a certain subset of experts could predict the future much better than chance and other experts, especially on shorter range forecasts

Conclusions (4)

- So ...
 - there are experts who are better than others at seeing the present and seeing the (near term) future
 - but **none of them** may be better than sophisticated algorithms, which are also far from accurate
- If relying on experts, what should we look for:
 - Previewing next month a bit: while the average expert did not do very well, the average of experts – or even non-experts – often does very well
 - So ... don't expect much from one expert but look for
 - Foxes of course, but how can you spot one?

Conclusions (5) and Questions

- Beware the seemingly compelling, long-winded, but ultimately one-sided and sometimes belligerent arguments; a staple of both a hedgehog AND unfortunately a person that the media loves to use
- Look for those who consider and articulate multiple underlying forces and outcomes, randomness, ambiguity, and uncertainty; who will discuss opposing views seriously and not dismissively, speak in probabilities rather than certainties, and are willing to admit they may be, and indeed have been, wrong
- Could we ourselves be better judges and forecasters if we emulated their traits? Certainly because most of us are already somewhere along the hedgehog-fox continuum
- Beware our own weaknesses, e.g., only seeking sources that validate our own beliefs

A Preview of November's BIF

Where else can we find guidance?

- You will just need to wait until next month
- But here is one hint, relying on a group of like thinking experts in a domain can be worse than a single expert in that field

But we will discuss

- The *Wisdom of Crowds* and similar sources
- Humans, and animals, effectively working in concert
- Formal structures for effective group decision making and predicting

A perhaps less dense (detail oriented) discussion than tonight?

If Time Permits – a Little More on the Study (1)

-
- Several methods used to assess cognitive style – identifying the foxes vs. the hedgehogs, including these 3 (three)
 - (1) Capture of free-flowing thoughts – what did they consider, what was the ratio of supporting ideas vs. opposing ideas in their thinking
 - Are causal connections (events, people, ripple effects) considered piecemeal or systematically?
 - Do they recognize that decision makers must make tradeoffs & compromises?
 - Do they recognize that sensible people can disagree on the same problem and consider why divergent perceptions exist?

If Time Permits – a Little More on the Study (1)

-
- (2) Bayesian belief-updating exercises – how did they react to confirmation / disconfirmation of their predictions? (154 experts from the regional forecasting exercises)
 - Ex ante process:
 - Query – how confident are you of your assessment of the underlying forces shaping events in [x]?
 - Query – how likely to be correct is the most influential position contrary to your position? (allowing for multiple alternatives to be considered)
 - Query – assuming your assessment of these conditions is correct, rank order the following scenarios in likelihood to occur
 - Query – assuming your assessment is incorrect, again rank order the alternatives scenarios
 - Query – taking all of the above into consideration, what is your best bottom-line probability estimate for each future?

-

If Time Permits – a Little More on the Study (3)

-
- Ex Post process (61-90% reinterview rate):
 - First – tested ability to recall prior forecast(s)
 - Second – (gently) remind them of what they actually did predict
 - Third – a “Taking Stock” questionnaire – rate their agreement / disagreement with 10 statements:
 - Presumption that same underlying forces would stay in effect – failed
 - Presumption of no external shocks would occur – failed
 - You appreciate how close we came to an outcome different than what did occur
 - You appreciate how easily a different outcome might still occur
 - Forecasting can produce valuable insights into validity of competing viewpoints
 - Politics is inherently unpredictable
 - Forecasting exercises are profoundly misleading assigning too much credit to lucky winners and blame to unlucky losers
 - Looking back, my assessment of underlying forces shaping events was sound
 - Looking back, my conceptual framework or theoretical principles were sound
 - Looking back, it was a good idea to overestimate some and underestimate other probabilities

If Time Permits – a Little More on the Study (4)

-
- Finally, “posterior probability questions”, the experts were:
 - Permitted to alter their initial probability predictions – if they chose to do so
 - Asked to assess their confidence in:
 - their initial assessment of underlying factors
 - the correctness of the competing points of view that they previously id’d